

Residual vascular risk despite glycemic control: dual-cohort + Mendelian randomization

ARM 1 — Discovery
HRS 2016 VBS (USA)
N = 3,686 with HbA1c
+ NT-proBNP

Exposure: In NT-proBNP
Outcome: all-cause death
(N=523 DM-controlled;
98 deaths, 2016–2023)

ARM 2 — Extension
CHARLS 2015 (China)
N = 10,959 with HbA1c
+ hsCRP

Exposure: In hsCRP
Outcome: incident CVD
(N=381 DM-controlled;
113 events, 2015–2020)

ARM 3 — Triangulation
Two-sample MR
(OpenGWAS)

Instruments: NPPA/NPPB cis
(rs198389, F=180; rs5068)
Outcomes: CAD, MI, HF,
stroke (6 consortia)

Analyses: survey-weighted & cluster-robust Cox models (12 covariates incl. cystatin C)
MICE m=20 · bootstrap · restricted cubic splines · competing-risk CIF · QBA · E-values

Observational arms
NT-proBNP and hsCRP identify
residual risk in HbA1c-controlled DM
HR 1.61 (1.10–2.34) death; 1.96 (1.20–3.21) CVD

Genetic arm
Marker-driver dissociation:
NT-proBNP null for CAD/MI/HF;
NPPA pathway protective for stroke
(IVW OR 0.91; rs5068 OR 0.63)